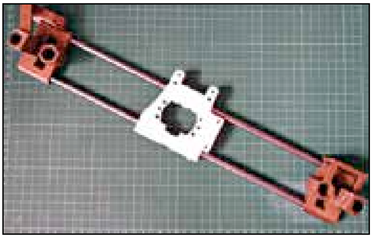
It's available for pre-order on their website and from their videos, seems like a lot of fun.



The X carriage mounted on the X axis

The X Carriage

The extruder and its motor are mounted on a part called the X Carriage (yes, it's printable too). The X Carriage runs travels across the X axis on smooth rails and is powered by a belt driven stepper motor.

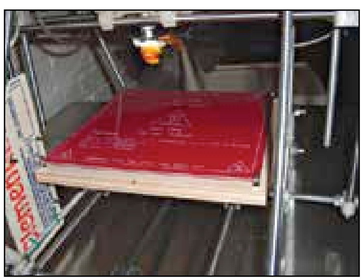
The entire X axis comprises of two X end idlers which are non-identical (printed parts again). These parts contain slots for the entire assembly to move across the Z axis using belt driven stepper motors mounted on the top two vertices and the stepper motor that controls the X axis is mounted on the left X end idler.

The Print Plate

The print plate is a two part structure and is basically two flat surfaces stacked one on top of another. While the bottom surface is mounted on the rails and runs back and forth, the actual printing surface is mounted on top. The Y axis is controlled by a stepper motor and a belt drive as well. Needless to say, both surfaces must be completely level.

The upper surface is where the printing actually happens and is mounted using screws which are placed inside springs. The springs help level the printing surface.

Because the print surface cools over the course of the print, the chances of the model warping, cracking and shattering very much exist. This is because the inner and outer sides cool at different level. As the outer side can cool faster than the inner side which has the advantage of being able to stay insulated and keep the heat in, the model warps and is rendered useless.



Heated Print Plate

This can be avoided by using a heated print plate. These print plates can go as high as 100 degrees Celsius and needless to say, be warned and